



TobiasBelch

AI ENGINEER

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about

AI Engineer with 17 years of software engineering experience, the last 2 focused on integrating LLMs and AI agents into production healthcare platforms. Bridges AI capabilities with robust full-stack delivery — owns AI agent orchestration, token optimisation, prompt engineering, and data pipeline architecture. Combines deep TypeScript, React, and Node.js expertise with real-time data streaming and visualisation skills built across regulated health, e-commerce, and sustainability domains. Sought out as a technical reference for AI initiatives, stakeholder communication, and mentoring engineers in best practices.

skills

AI / ML

LLM Integration (OpenAI, Anthropic)
AI Agent Orchestration
Prompt Engineering & Optimisation
AI-Assisted UI Generation
Token Usage & Reliability Tuning

Data & Streaming

Apache Flink / Apache NiFi
Real-Time Data Pipelines
AMQP / Message Brokers
D3 / Data Visualisation
Data Modelling & Simulation

Engineering

TypeScript / JavaScript
React / Next.js / Node.js
REST / GraphQL API Design
NestJS / BFF Architecture
Docker / CI/CD
Architecture Decision Records

work experience

Senior Software Engineer – DocMorris

Berlin, Germany | 08/2023 – Present

Digital healthcare and online pharmacy platform in the German market.

- Led AI agent initiatives across the engineering org — researched token usage, reliability, and response predictability for LLM-powered features.
- Designed and implemented a cross-platform Design System structured explicitly for AI-assisted UI generation, enabling rapid component creation via LLM prompting.
- Architected a white-label platform migration with 3 brands onboarded; drove frontend and BFF architecture evolution.
- Advised on AI integration strategy, evaluating model providers, prompt patterns, and cost/quality trade-offs for production use cases.
- Mentored 5 engineers on TypeScript, React, and AI-adjacent engineering practices.

work experience

Head of Engineering – Made2Flow

Berlin, Germany | 12/2022 – 07/2023

Sustainability platform for fashion supply-chain decision support.

- Designed a next-generation domain data model from scratch with the CPO, replacing a fragmented model insufficient to support data-intensive product use cases.
- Built and led a team of 4 engineers, including hiring 2; owned technical partner communication and integration planning.
- Evaluated data pipeline and analytics architecture to support decision-support features.

Technical Lead – eHealth-Tec GmbH

Berlin, Germany | 04/2020 – 11/2022

Digital health products for patients, doctors, and pharmacists.

- Built and owned the frontend and BFF layers of a white-label online pharmacy platform serving 2 clients in production (TypeScript, React, Next.js, NestJS).
- Designed Backend-for-Frontend layers as anti-corruption boundaries, shaping API contracts for downstream data consumers.
- Introduced a requirements-driven architecture process — ADRs, design reviews, and structured refinement — laying groundwork for data-informed technical decisions.
- Delivered a gematik-regulated browser-based e-prescription tool, navigating strict compliance and data privacy requirements.

Head of Engineering – Medtower UG

Berlin, Germany | 09/2018 – 04/2020

Software and hardware solutions for hospital operations and visibility.

- Owned architecture of a real-time IoT platform — microservices, SPA, and live device tracking (Node.js, React, TypeScript).
- Designed a multi-device real-time communication system using AMQP 1.0 and Apache Qpid for streaming device telemetry.
- Built 3D indoor maps with live device location and status visualisation in an offline environment (OpenMapTiles, D3).
- Managed external Apache NiFi consultants for ETL data pipeline delivery.

earlier experience

Software Developer – remind me GmbH

02/2018 – 08/2018

Authored the technical concept for a white-label product transformation. Implemented feature-toggle architecture with lazy loading for flexible multi-context deployment. Mentored developers in a Kanban team of 5.

Software Developer – Cedavis Technology GmbH

08/2009 – 01/2018

Evolved frontend architecture over 9 years — introduced Angular 1, then migrated to React and TypeScript. Built a custom D3 chart component library from scratch. Implemented data simulation and forecasting features for healthcare capacity planning. Engineered a PDF reporting module using Java Nashorn to render React components server-side.

languages

German (Native) · English (CEFR C-level)

education

Berlin University of Applied Sciences (BHT)

2007 – 2011

Bachelor of Science, Computer Science

certifications

iSAQB Certified Professional for Software Architecture – Foundation

05/2022

oose

Domain Driven Design

12/2021

entwickler.de